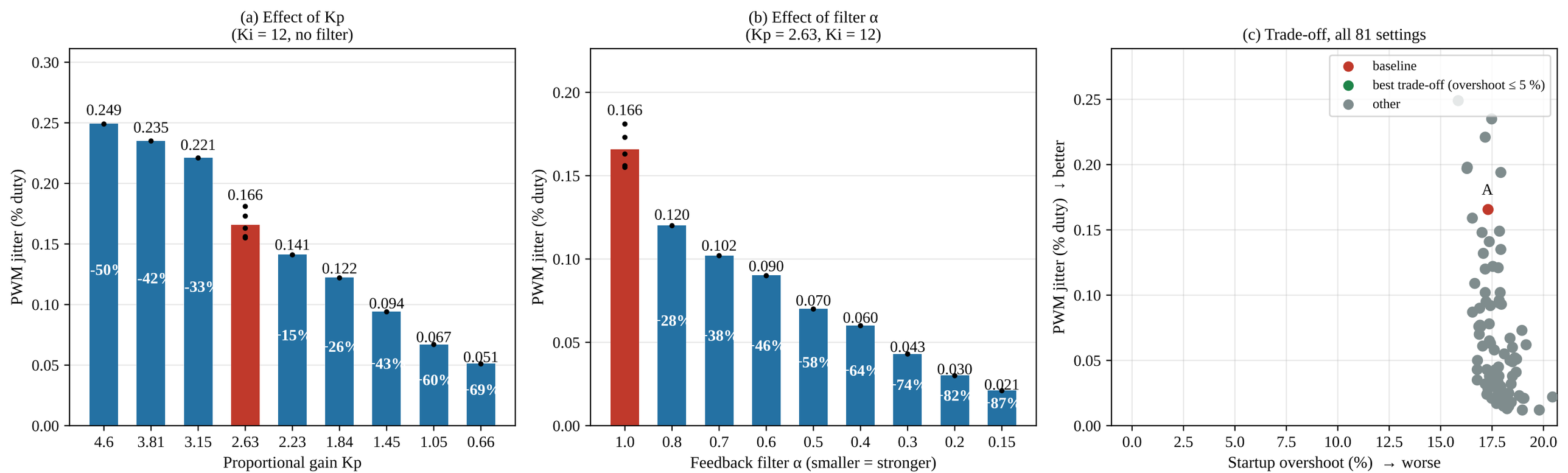
(a) Effect of Kp
(Ki = 12, no filter)
PWM jitter (% duty)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.249
0.235
0.221
0.166
0.141
0.122
0.094
0.067
0.051
-50%
-42%
-33%
-15%
-26%
-43%
-60%
-69%
4.6
3.81
3.15
2.63
2.23
1.84
1.45
1.05
0.66
Proportional gain Kp
(b) Effect of filter α
(Kp = 2.63, Ki = 12)
PWM jitter (% duty)
0.20
0.15
0.10
0.05
0.00
0.166
0.120
0.102
0.090
0.070
0.060
0.043
0.030
0.021
-28%
-38%
-46%
-58%
-64%
-74%
-82%
-87%
1.0
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.15
Feedback filter α (smaller = stronger)
(c) Trade-off, all 81 settings
baseline
best trade-off (overshoot ≤ 5 %)
other
A
PWM jitter (% duty) ↓ better
0.25
0.20
0.15
0.10
0.05
0.00
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Startup overshoot (%) → worse